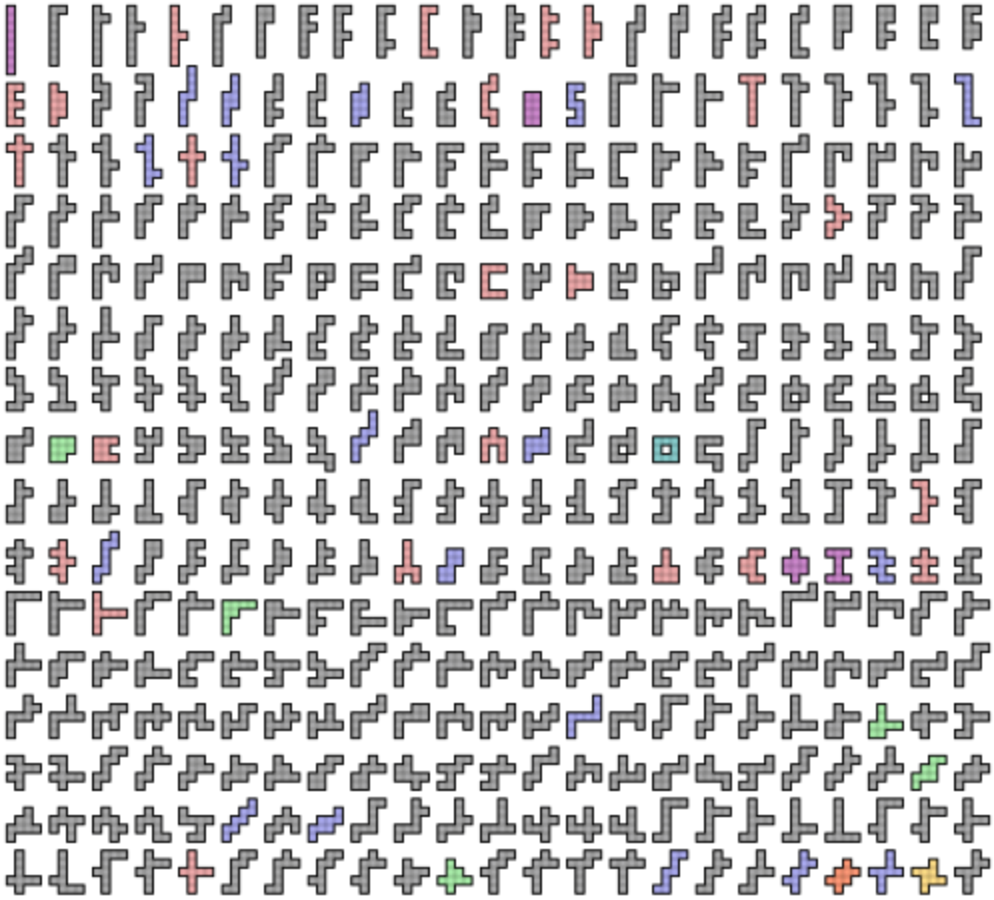
Not to be confused with Otomino.



The 369 free octominoes

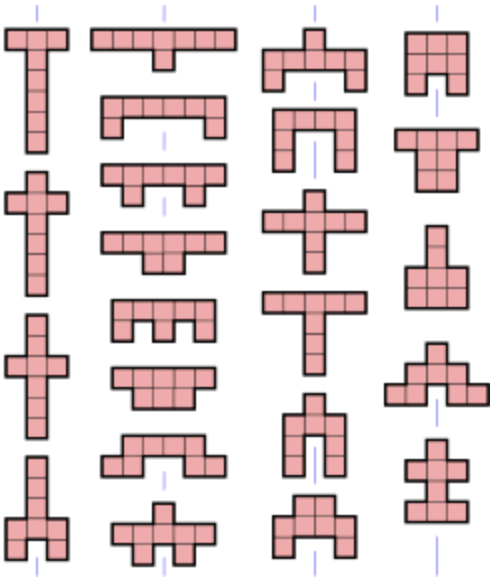
An **octomino** (or **8-omino**) is a polyomino of order 8; that is, a polygon in the plane made of 8 equal-sized squares connected edge to edge.1(https://en.wikipedia.org/wiki/Octomino#cite_note-1) When rotations and reflections are not considered to be distinct shapes, there are 369 different *free* octominoes. When reflections are considered distinct, there are 704 *one-sided* octominoes. When rotations are also considered distinct, there are 2,725 *fixed* octominoes.2(https://en.wikipedia.org/wiki/Octomino#cite_note-2)[[3]](https://en.wikipedia.org/wiki/Octomino#cite_note-Redelmeier-3)

Symmetry

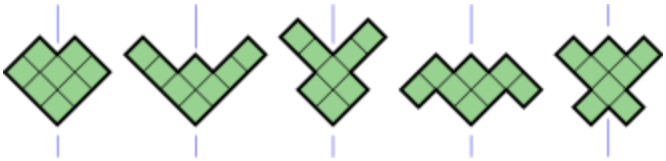
The figure shows all possible free octominoes, coloured according to their symmetry groups:

- 316 octominoes (coloured grey) have no symmetry. Their symmetry group consists only of the identity mapping.
- 23 octominoes (coloured red) have an axis of reflection symmetry aligned with the gridlines. Their symmetry group has two elements, the identity and the reflection in a line parallel to

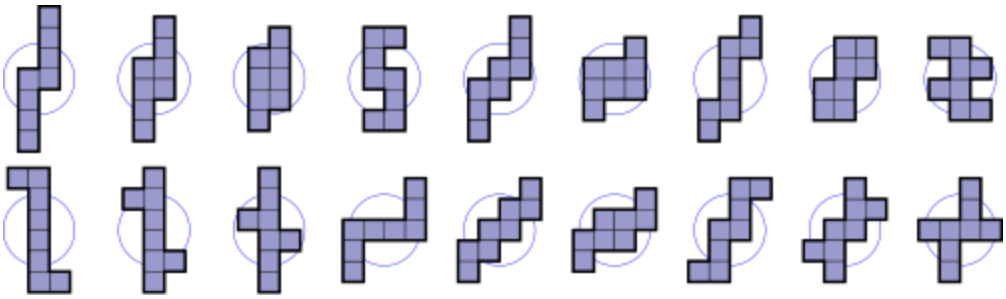
the sides of the squares.



- 5 octominoes (coloured green) have an axis of reflection symmetry at 45° to the gridlines. Their symmetry group has two elements, the identity and a diagonal reflection.



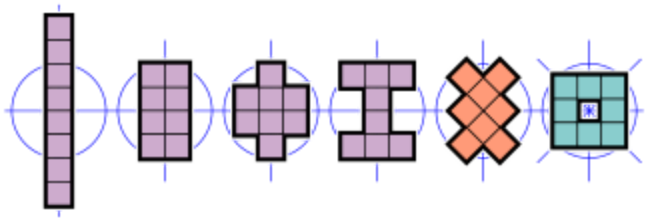
- 18 octominoes (coloured blue) have point symmetry, also known as rotational symmetry of order 2. Their symmetry group has two elements, the identity and the 180° rotation.



- 1 octomino (coloured yellow) has rotational symmetry of order 4. Its symmetry group has four elements, the identity and the 90°, 180° and 270° rotations.

- 4 octominoes (coloured purple) have two axes of reflection symmetry, both aligned with the gridlines. Their symmetry group has four elements, the identity, two reflections and the 180° rotation. It is the dihedral group of order 2, also known as the Klein four-group.

- 1 octomino (coloured orange) has two axes of reflection symmetry, both aligned with the diagonals. Its symmetry group is also the dihedral group of order 2 with four elements.
- 1 octomino (coloured cyan) has four axes of reflection symmetry, aligned with the gridlines and the diagonals, and rotational symmetry of order 4. Its symmetry group, the dihedral group of order 4, has eight elements.



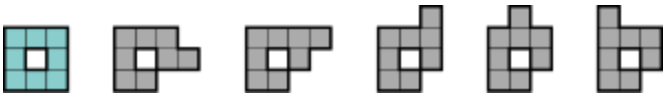
The set of octominoes is the lowest polyomino set in which all eight possible symmetries are realized. The next higher set with this property is the dodecomino (12-omino) set.3(https://en.wikipedia.org/wiki/Octomino#cite_note-Redelmeier-3)

If reflections of an octomino are considered distinct, as they are with one-sided octominoes, then the first, fourth and fifth categories above double in size, resulting in an extra 335 octominoes for a total of 704. If rotations are also considered distinct, then the octominoes from the first category count eightfold, the ones from the next three categories count fourfold, the ones from categories five to seven count twice, and the last octomino counts only once. This results in 316 × 8 + (23+5+18) × 4 + (1+4+1) × 2 + 1 = 2,725 fixed octominoes.

Packing and tiling

Of the 369 free octominoes, 320 satisfy the Conway criterion and 23 more can form a patch satisfying the criterion.4(https://en.wikipedia.org/wiki/Octomino#cite_note-4) The other 26 octominoes (including the 6 with holes) are unable to tessellate the plane.5(https://en.wikipedia.org/wiki/Octomino#cite_note-5)

Since 6 of the free octominoes have a hole, it is trivial to prove that the complete set of octominoes cannot be packed into a rectangle, and that not all octominoes can be tiled.



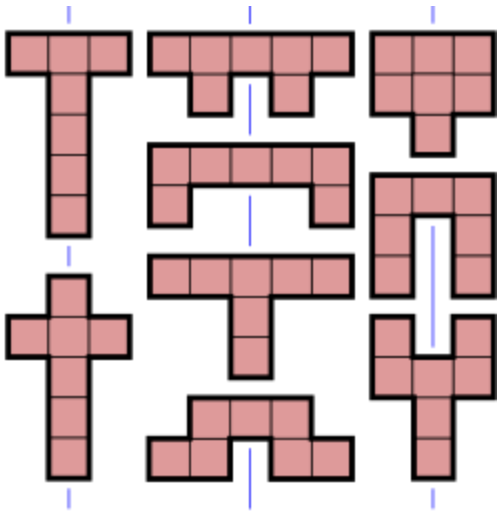
A **heptomino** (or **7-omino** or **septomino**) is a polyomino of order 7; that is, a polygon in the plane made of 7 equal-sized squares connected edge to edge.1(https://en.wikipedia.org/wiki/Heptomino#cite_note-1) The name of this type is formed with the prefix hept(a)-. When rotations and reflections are not considered to be distinct shapes, there are 108 different *free* heptominoes. When reflections are considered distinct, there

are 196 *one-sided* heptominoes. When rotations are also considered distinct, there are 760 *fixed* heptominoes.2(https://en.wikipedia.org/wiki/Heptomino#cite_note-2)[[3]](https://en.wikipedia.org/wiki/Heptomino#cite_note-3)

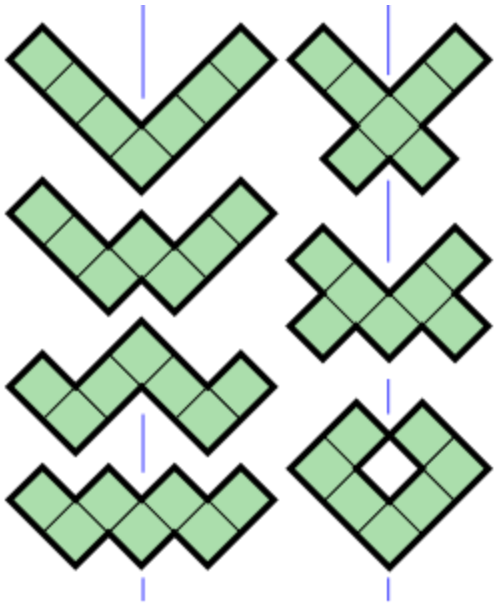
Symmetry

The figure shows all possible free heptominoes, coloured according to their symmetry groups:

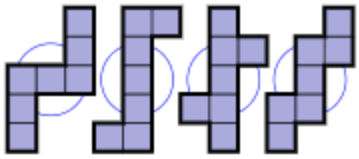
- 84 heptominoes (coloured grey) have no symmetry. Their symmetry group consists only of the identity mapping.
- 9 heptominoes (coloured red) have an axis of reflection symmetry aligned with the gridlines. Their symmetry group has two elements, the identity and the reflection in a line parallel to the sides of the squares.



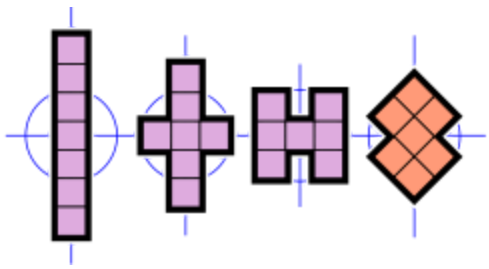
- 7 heptominoes (coloured green) have an axis of reflection symmetry at 45° to the gridlines. Their symmetry group has two elements, the identity and a diagonal reflection.



- 4 heptominoes (coloured blue) have point symmetry, also known as rotational symmetry of order 2. Their symmetry group has two elements, the identity and the 180° rotation.



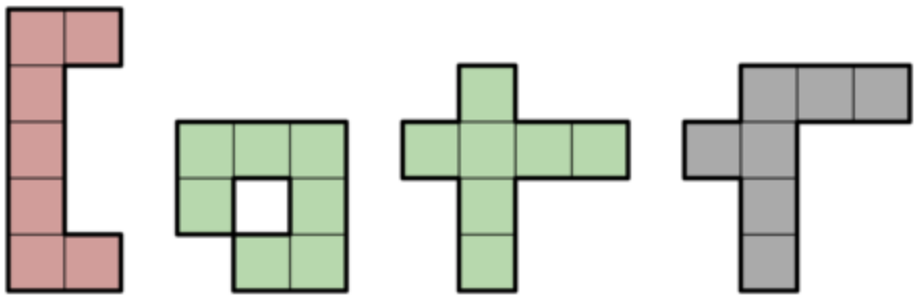
- 3 heptominoes (coloured purple) have two axes of reflection symmetry, both aligned with the gridlines. Their symmetry group has four elements, the identity, two reflections and the 180° rotation. It is the dihedral group of order 2, also known as the Klein four-group.
- 1 heptomino (coloured orange) has two axes of reflection symmetry, both aligned with the diagonals. Its symmetry group also has four elements. Its symmetry group is also the dihedral group of order 2 with four elements.



If reflections of a heptomino are considered distinct, as they are with one-sided heptominoes, then the first and fourth categories above would each double in size, resulting in an extra 88 heptominoes for a total of 196. If rotations are also considered distinct, then the heptominoes from the first category count eightfold, the ones from the next three categories count fourfold, and the ones from the last two categories count twice. This results in 84 × 8 + (9+7+4) × 4 + (3+1) × 2 = 760 fixed heptominoes.

Packing and tiling

Of the 108 free heptominoes, 101 satisfy the Conway criterion and 3 more can form a patch satisfying the criterion. Only 4 heptominoes fail to satisfy the criterion and these 4 are unable to tessellate the plane.4(https://en.wikipedia.org/wiki/Heptomino#cite_note-4)



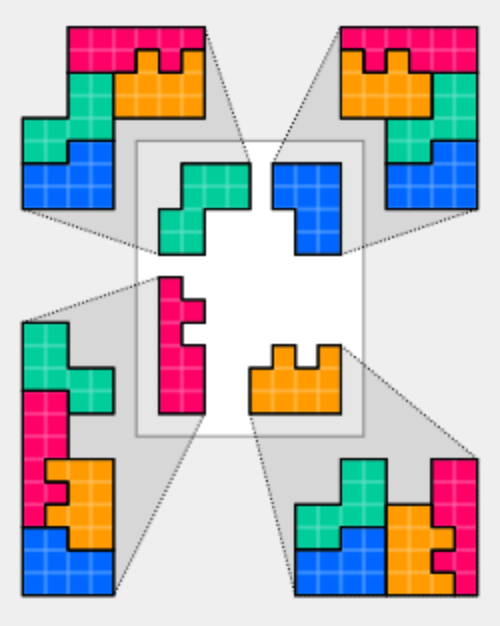
The four heptominoes incapable of tiling a plane, including the one heptomino with a hole

Although a complete set of the 108 free heptominoes has a total of 756 squares, it is not possible to tile a rectangle with that set. The proof of this is trivial, since there is one heptomino which has a hole.5(https://en.wikipedia.org/wiki/Heptomino#cite_note-Gruenbaum-5) It is also impossible to form a 757-square rectangle because 757 is a prime number.

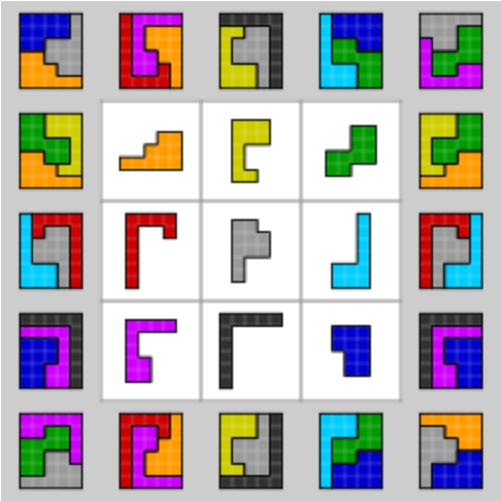
However, the set of 107 simply connected free heptominoes—that is, the ones without the hole—can tile a 7 by 107 (749-square) rectangle.6(https://en.wikipedia.org/wiki/Heptomino#cite_note-6) Furthermore, the complete set of free heptominoes can tile three 11-by-23 (253-square) rectangles, each with a one-square hole in the center; the complete set can also tile twelve 8 by 8 (64-square) squares with a one-square hole in the "center".7(https://en.wikipedia.org/wiki/Heptomino#cite_note-7)

Unlike both octominoes and nonominoes, no decomino has rotational symmetry of order 4.

Packing and tiling



A self-tiling tile set consisting of decominoes



A geometric magic square consisting of decominoes

195 decominoes have holes. This makes it trivial to prove that the complete set of decominoes cannot be packed into a rectangle, and that not all decominoes can be tiled.

The 4,460 decominos without holes comprise 44,600 unit squares. Thus, the largest square that can be tiled with distinct decominoes is at most 210 units on a side (210 squared is 44,100). Such a square containing 4,410 decominoes was constructed by Livio Zucca.3(https://en.wikipedia.org/wiki/Decomino#cite_note-3)

From Wikipedia, the free encyclopedia

A nonomino or Jigsaw Sudoku puzzle, as seen in *The Sunday Telegraph*

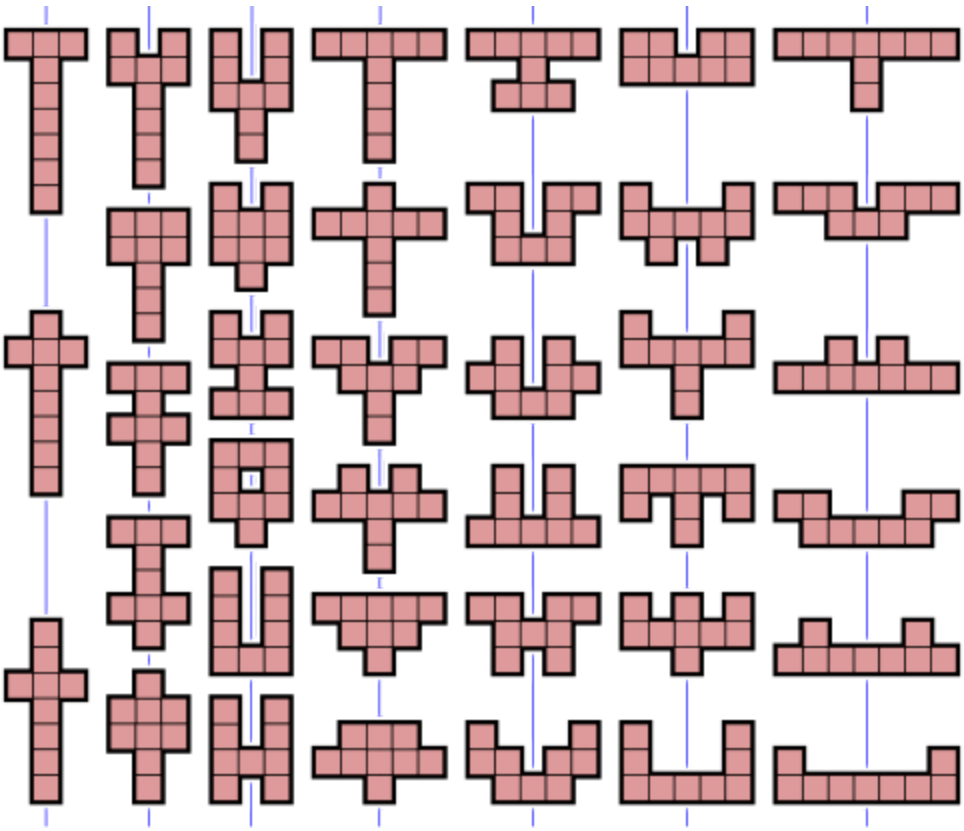
A **nonomino** (or **enneomino** or **9-omino**) is a polyomino of order 9; that is, a polygon in the plane made of 9 equal-sized squares connected edge to edge.1(https://en.wikipedia.org/wiki/Nonomino#cite_note-1) The name of this type of figure is formed with the prefix non(a)-. When rotations and reflections are not considered to be distinct

shapes, there are 1,285 different *free* nonominoes. When reflections are considered distinct, there are 2,500 *one-sided* nonominoes. When rotations are also considered distinct, there are 9,910 *fixed* nonominoes.2(https://en.wikipedia.org/wiki/Nonomino#cite_note-Redelmeier-2)

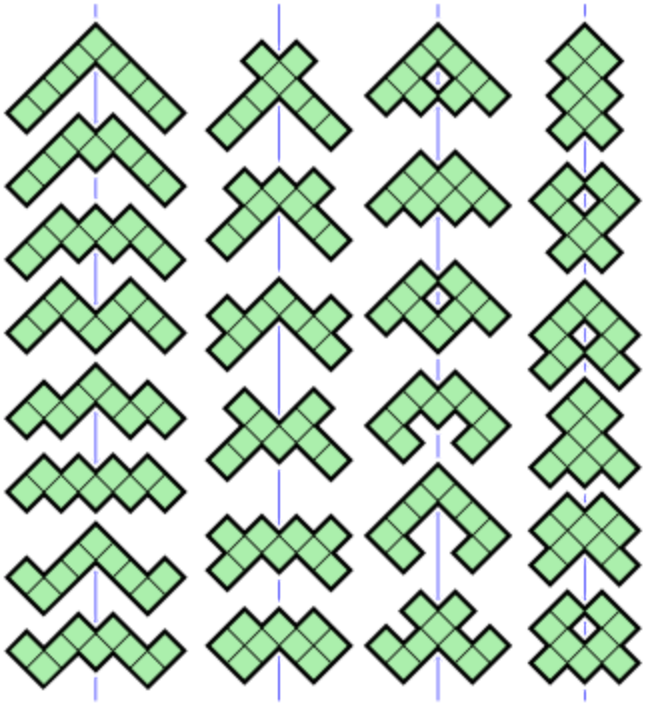
Symmetry

The 1,285 free nonominoes can be classified according to their symmetry groups:2(https://en.wikipedia.org/wiki/Nonomino#cite_note-Redelmeier-2)

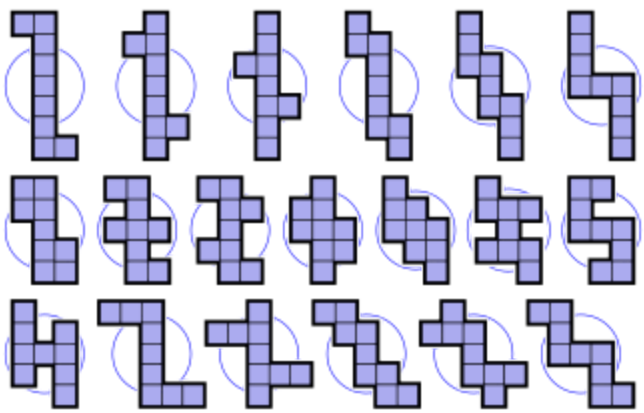
- 1,196 nonominoes have no symmetry. Their symmetry group consists only of the identity mapping.
- 38 nonominoes have an axis of reflection symmetry aligned with the gridlines. Their symmetry group has two elements, the identity and the reflection in a line parallel to the sides of the squares.



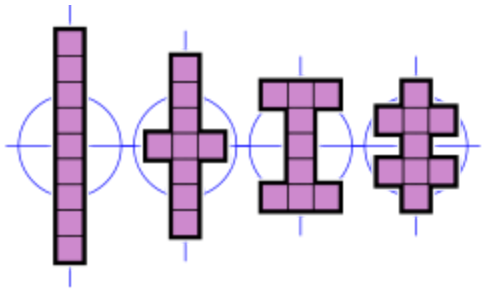
- 26 nonominoes have an axis of reflection symmetry at 45° to the gridlines. Their symmetry group has two elements, the identity and a diagonal reflection.



- 19 nonominoes have point symmetry, also known as rotational symmetry of order 2. Their symmetry group has two elements, the identity and the 180° rotation.

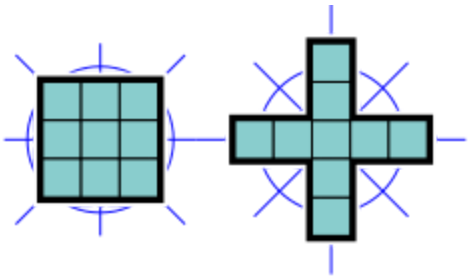


- 4 nonominoes have two axes of reflection symmetry, both aligned with the gridlines. Their symmetry group has four elements, the identity, two reflections and the 180° rotation. It is the dihedral group of order 2, also known as the Klein four-group.



- 2 nonominoes have four axes of reflection symmetry, aligned with the gridlines and the diagonals, and rotational symmetry of order 4. Their symmetry group, the dihedral group of

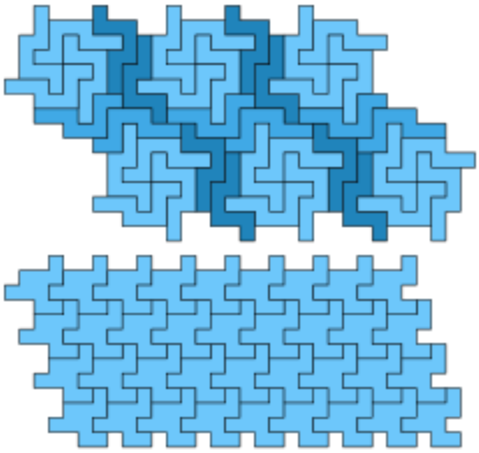
order 4, has eight elements.



Unlike octominoes, there are no nonominoes with rotational symmetry of order 4 or with two axes of reflection symmetry aligned with the diagonals.

If reflections of a nonomino are considered distinct, as they are with one-sided nonominoes, then the first and fourth categories above double in size, resulting in an extra 1,215 nonominoes for a total of 2,500. If rotations are also considered distinct, then the nonominoes from the first category count eightfold, the ones from the next three categories count fourfold, the ones from the fifth category count twice, and the ones from the last category count only once. This results in 1,196 × 8 + (38+26+19) × 4 + 4 × 2 + 2 = 9,910 fixed nonominoes.

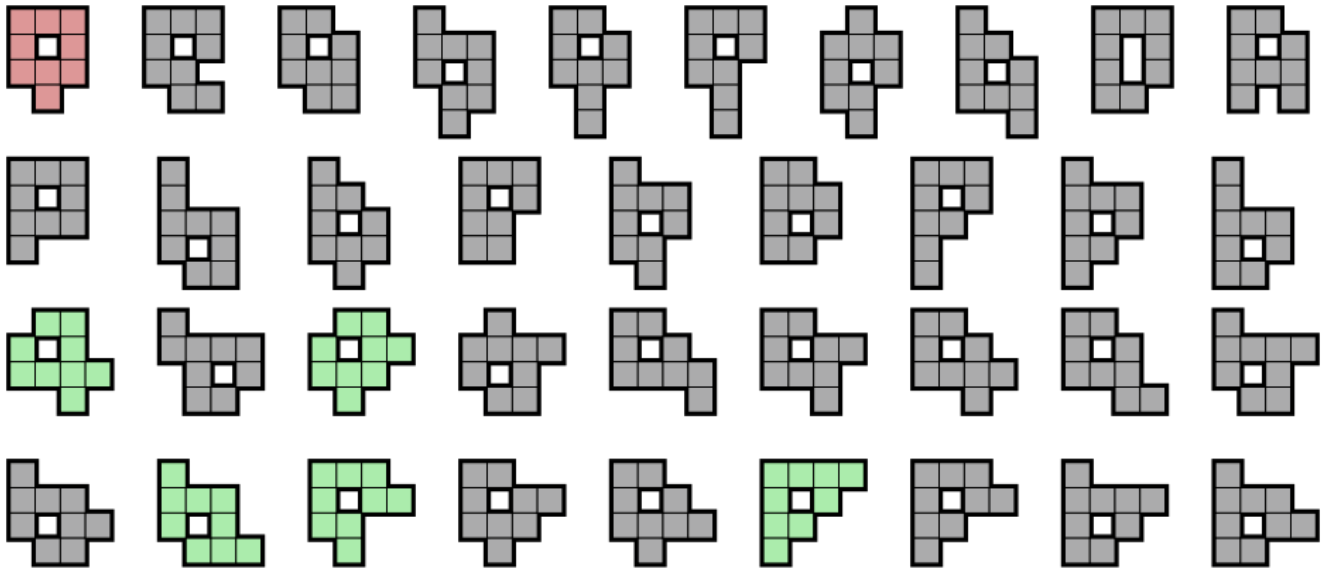
Packing and tiling



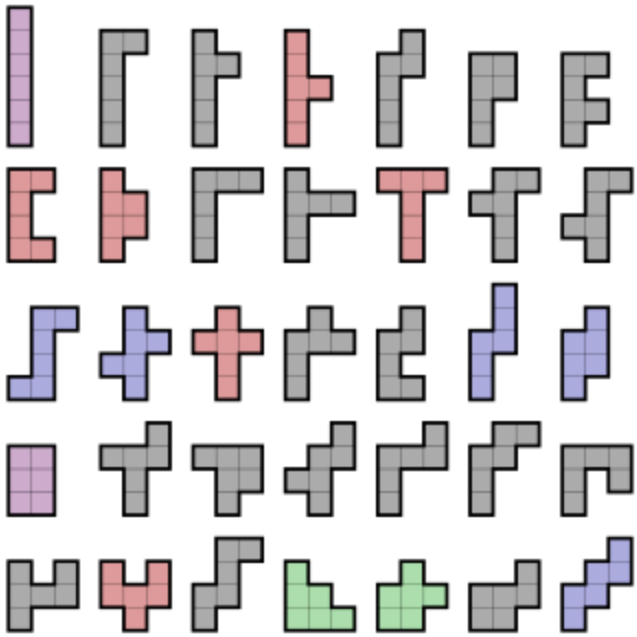
The two nonominoes which can tile the plane, but cannot form a patch which satisfies the Conway criterion.

37 nonominoes have holes.3(https://en.wikipedia.org/wiki/Nonomino#cite_note-3)[[4]](https://en.wikipedia.org/wiki/Nonomino#cite_note-4) Therefore a complete set cannot be packed into a rectangle and not all nonominoes have tilings. Of the 1285 free nonominoes, 960 satisfy the Conway criterion and 88 more can form a patch satisfying the criterion. Two additional nonominoes admit tilings, but satisfy neither of the previous criteria.5(https://en.wikipedia.org/wiki/Nonomino#cite_note-5) This is the lowest order of polyomino for which such exceptions exist.6(https://en.wikipedia.org/wiki/Nonomino#cite_note-6)

One nonomino has a two-square hole (second rightmost in the top row) and is the smallest polyomino with such a hole.



From Wikipedia, the free encyclopedia



The 35 free hexominoes

A **hexomino** (or **6-omino**) is a polyomino of order 6; that is, a polygon in the plane made of 6 equal-sized squares connected edge to edge.1(https://en.wikipedia.org/wiki/Hexomino#cite_note-1) The name of this type of figure is formed with the prefix hex(a)-. When rotations and reflections are not considered to be distinct shapes, there are 35 different *free* hexominoes. When reflections are considered distinct, there

are 60 *one-sided* hexominoes. When rotations are also considered distinct, there are 216 *fixed* hexominoes.2(https://en.wikipedia.org/wiki/Hexomino#cite_note-2)[[3]](https://en.wikipedia.org/wiki/Hexomino#cite_note-3)

Symmetry

The figure above shows all 35 possible free hexominoes, coloured according to their symmetry groups:

- The twenty grey hexominoes have no symmetry. Their symmetry group consists only of the identity mapping.
- The six red hexominoes have an axis of mirror symmetry parallel to the gridlines. Their symmetry group has two elements, the identity and a reflection in a line parallel to the sides of the squares.
- The two green hexominoes have an axis of mirror symmetry at 45° to the gridlines. Their symmetry group has two elements, the identity and a diagonal reflection.
- The five blue hexominoes have point symmetry, also known as rotational symmetry of order 2. Their symmetry group has two elements, the identity and the 180° rotation.
- The two purple hexominoes have two axes of mirror symmetry, both parallel to the gridlines (thus one horizontal axis and one vertical axis). Their symmetry group has four elements. It is the dihedral group of order 2, also known as the Klein four-group.

If reflections of a hexomino are considered distinct, as they are with one-sided hexominoes, then the first and fourth categories above would each double in size, resulting in an extra 25 hexominoes for a total of 60. If rotations are also considered distinct, then the hexominoes from the first category count eightfold, the ones from the next three categories count fourfold, and the ones from the last category count twice. This results in $20 \times 8 + (6 + 2 + 5) \times 4 + 2 \times 2 = 216$ fixed hexominoes.

Packing and tiling

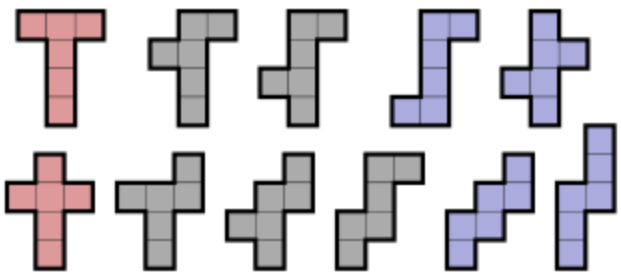
Each of the 35 hexominoes satisfies the Conway criterion; hence, every hexomino is capable of tiling the plane.4(https://en.wikipedia.org/wiki/Hexomino#cite_note-4)

Although a complete set of 35 hexominoes has a total of 210 squares, it is not possible to pack them into a rectangle. (Such an arrangement is possible with the 12 pentominoes, which can be packed into any of the rectangles 3×20, 4×15, 5×12 and 6×10.) A simple way to demonstrate that such a packing of hexominoes is not possible is via a parity argument. If the hexominoes are placed on a checkerboard pattern, then 11 of the hexominoes will cover an even number of black squares (either 2 white and 4 black or vice versa) and the other 24 hexominoes will cover an odd number of black squares (3 white and 3 black). Overall, an even number of black squares will be covered in any arrangement. However, any rectangle of 210

squares will have 105 black squares and 105 white squares, and therefore cannot be covered by the 35 hexominoes.

However, there are other simple figures of 210 squares that can be packed with the hexominoes. For example, a 15 × 15 square with a 3 × 5 rectangle removed from the centre has 210 squares. With checkerboard colouring, it has 106 white and 104 black squares (or vice versa), so parity does not prevent a packing, and a packing is indeed possible.5(https://en.wikipedia.org/wiki/Hexomino#cite_note-5) It is also possible for two sets of pieces to fit a rectangle of size 420, or for the set of 60 one-sided hexominoes (18 of which cover an even number of black squares) to fit a rectangle of size 360.6(https://en.wikipedia.org/wiki/Hexomino#cite_note-6)

Polyhedral nets for the cube

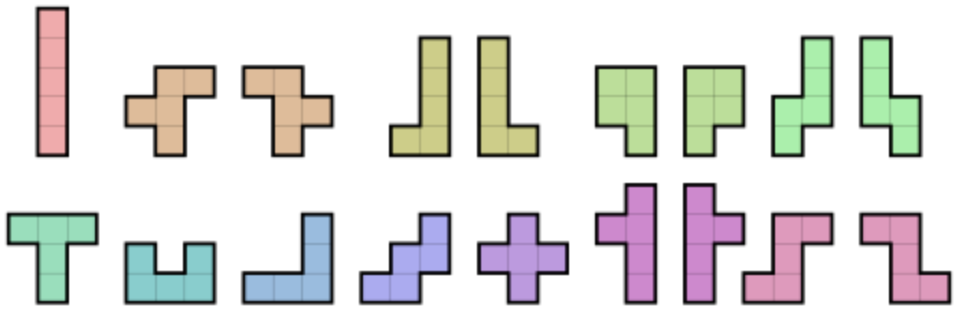


All 11 unfoldings of the cube

A polyhedral net for the cube is necessarily a hexomino, with 11 hexominoes (shown at right) actually being nets. They appear on the right, again coloured according to their symmetry groups.

A polyhedral net for the cube cannot contain the O-tetromino, nor the I-pentomino, the U-pentomino, or the V-pentomino.

From Wikipedia, the free encyclopedia



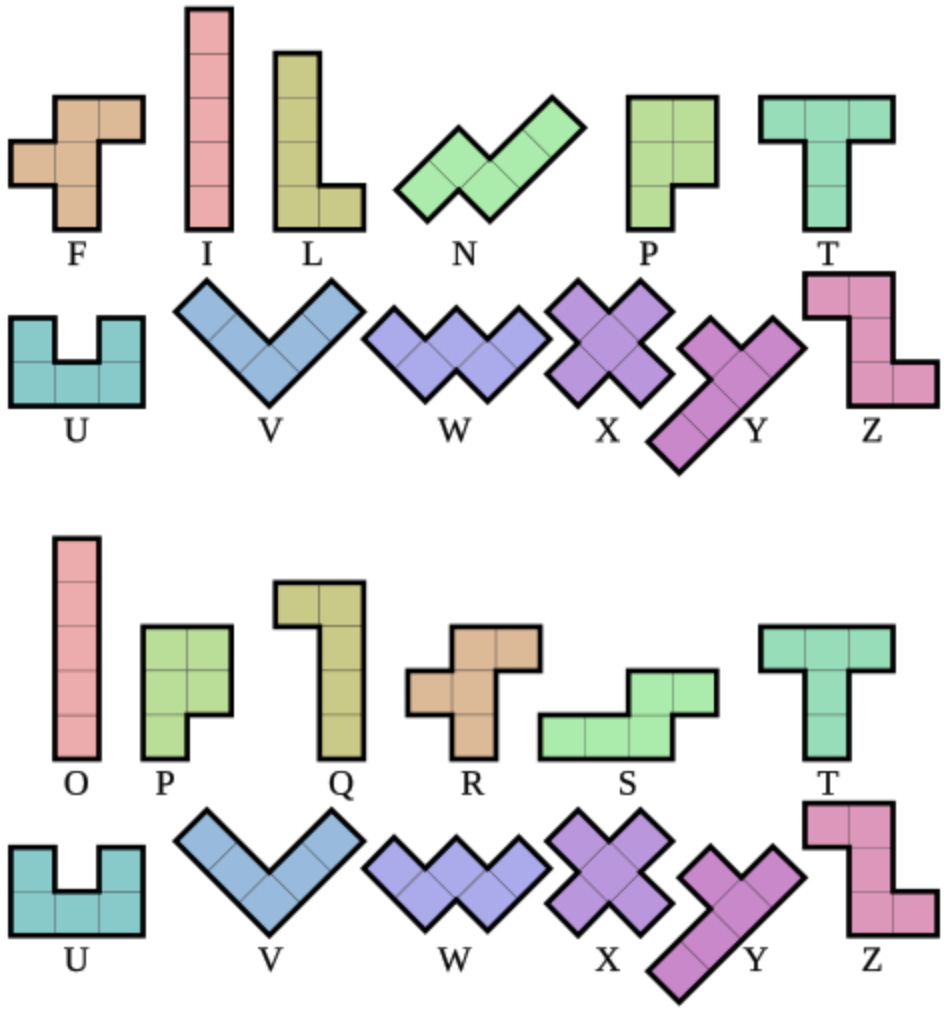
The 12 pentominoes can form 18 different shapes, with 6 of them (the chiral pentominoes) being mirrored.

A **pentomino** (or **5-omino**) is a polyomino of order 5; that is, a polygon in the plane made of 5 equal-sized squares connected edge to edge. The term is derived from the Greek word for '5' and "domino". When rotations and reflections are not considered to be distinct shapes, there are 12 different *free* pentominoes. When reflections are considered distinct, there are 18 *one-sided* pentominoes. When rotations are also considered distinct, there are 63 *fixed* pentominoes.

Pentomino tiling puzzles and games are popular in recreational mathematics.1(https://en.wikipedia.org/wiki/Pentomino#cite_note-Harshbarger-1) Usually, video games such as *Tetris* imitations and *Rampart* consider mirror reflections to be distinct, and thus use the full set of 18 one-sided pentominoes. (Tetris itself uses 4-square shapes called tetrominos.)

Each of the twelve pentominoes satisfies the Conway criterion; hence, every pentomino is capable of tiling the plane.2(https://en.wikipedia.org/wiki/Pentomino#cite_note-2) Each chiral pentomino can tile the plane without being reflected.3(https://en.wikipedia.org/wiki/Pentomino#cite_note-3)

History



Comparison of labeling schemes for the 12 possible pentomino shapes. The first naming convention is the one used in this article. The second method is Conway's.

The earliest puzzle containing a complete set of pentominoes appeared in Henry Dudeney's book *The Canterbury Puzzles*, published in 1907.4(https://en.wikipedia.org/wiki/Pentomino#cite_note-4) The earliest tilings of rectangles with a complete set of pentominoes appeared in the Problemist Fairy Chess Supplement in 1935, and further tiling problems were explored in the PFCS, and its successor, the Fairy Chess Review.5(https://en.wikipedia.org/wiki/Pentomino#cite_note-5)[[6]] (https://en.wikipedia.org/wiki/Pentomino#cite_note-6):127

Pentominoes were formally defined by American professor Solomon W. Golomb starting in 1953 and later in his 1965 book *Polyominoes: Puzzles, Patterns, Problems, and Packings*.1(https://en.wikipedia.org/wiki/Pentomino#cite_note-Harshbarger-1)[[7]] (https://en.wikipedia.org/wiki/Pentomino#cite_note-7) They were introduced to the general public by Martin Gardner in his October 1965 Mathematical Games column in Scientific American. Golomb coined the term "pentomino" from the Ancient Greek πέντε / *pénte*, "five", and the -omino of domino, fancifully interpreting the "d-" of "domino" as if it were a form of the

Greek prefix "di-" (two). Golomb named the 12 *free* pentominoes after letters of the Latin alphabet that they resemble, using the mnemonic FILiPiNo along with the end of the alphabet (TUVWXYZ).8(https://en.wikipedia.org/wiki/Pentomino#cite_note-8):23

John Horton Conway proposed an alternate labeling scheme for pentominoes, using O instead of I, Q instead of L, R instead of F, and S instead of N. The resemblance to the letters is more strained, especially for the O pentomino, but this scheme has the advantage of using 12 consecutive letters of the alphabet. It is used by convention in discussing Conway's Game of Life, where, for example, one speaks of the R-pentomino instead of the F-pentomino.

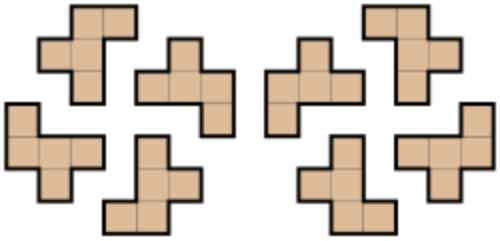
Symmetry

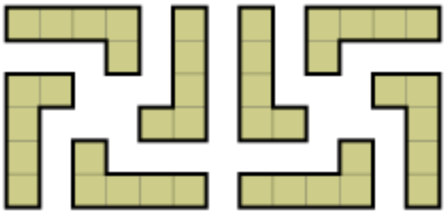
- F, L, N, P, and Y can be oriented in 8 ways: 4 by rotation, and 4 more for the mirror image. Their symmetry group consists only of the identity mapping.
- T, and U can be oriented in 4 ways by rotation. They have an axis of reflection aligned with the gridlines. Their symmetry group has two elements, the identity and the reflection in a line parallel to the sides of the squares.
- V and W also can be oriented in 4 ways by rotation. They have an axis of reflection symmetry at 45° to the gridlines. Their symmetry group has two elements, the identity and a diagonal reflection.
- Z can be oriented in 4 ways: 2 by rotation, and 2 more for the mirror image. It has point symmetry, also known as rotational symmetry of order 2. Its symmetry group has two elements, the identity and the 180° rotation.
- I can be oriented in 2 ways by rotation. It has two axes of reflection symmetry, both aligned with the gridlines. Its symmetry group has four elements, the identity, two reflections and the 180° rotation. It is the dihedral group of order 2, also known as the Klein four-group.
- X can be oriented in only one way. It has four axes of reflection symmetry, aligned with the gridlines and the diagonals, and rotational symmetry of order 4. Its symmetry group, the dihedral group of order 4, has eight elements.

The F, L, N, P, Y, and Z pentominoes are chiral; adding their reflections (F′, J, N′, Q, Y′, S) brings the number of *one-sided* pentominoes to 18. If rotations are also considered distinct, then the pentominoes from the first category count eightfold, the ones from the next three categories (T, U, V, W, Z) count fourfold, I counts twice, and X counts only once. This results in 5×8 + 5×4 + 2 + 1 = 63 *fixed* pentominoes.

The eight possible orientations of the F, L, N, P, and Y pentominoes, and the four possible orientations of the T, U, V, W, and Z pentominoes are illustrated:

- Eight possible orientations

-  F-pentomino



-

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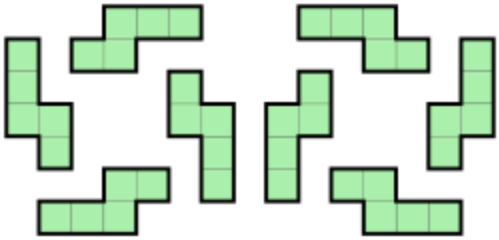
L-pentomino
```



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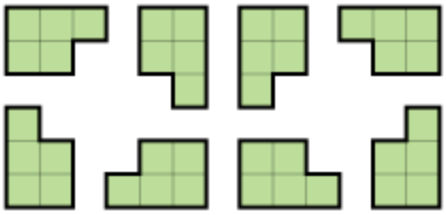
```
N-pentomino
```



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P-pentomino
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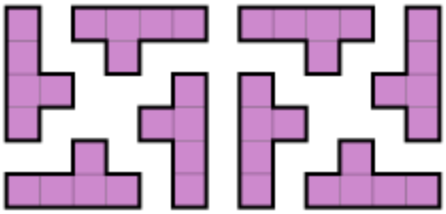

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Y-pentomino
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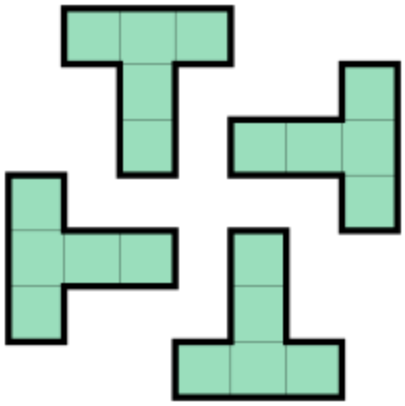
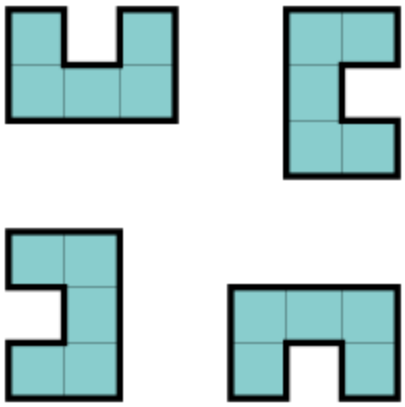
- Four possible orientations



- T-pentomino



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U-pentomino
```
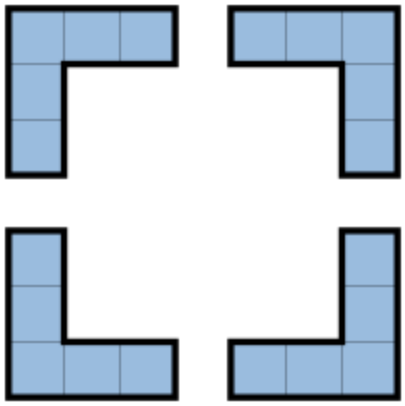
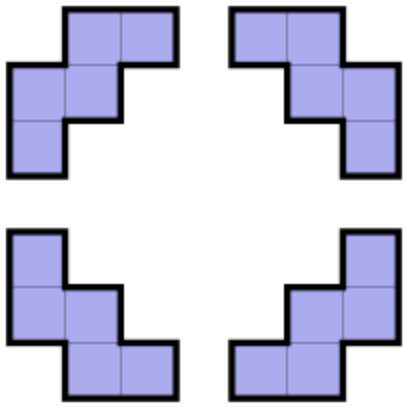



-

```
V-pentomino
```



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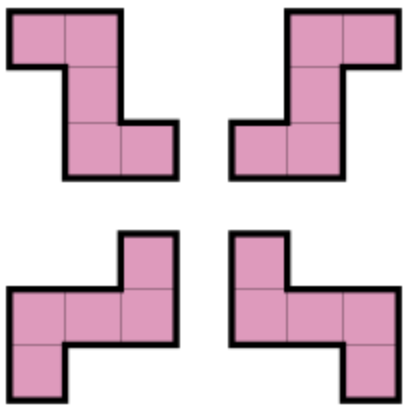
```
W-pentomino
```



-

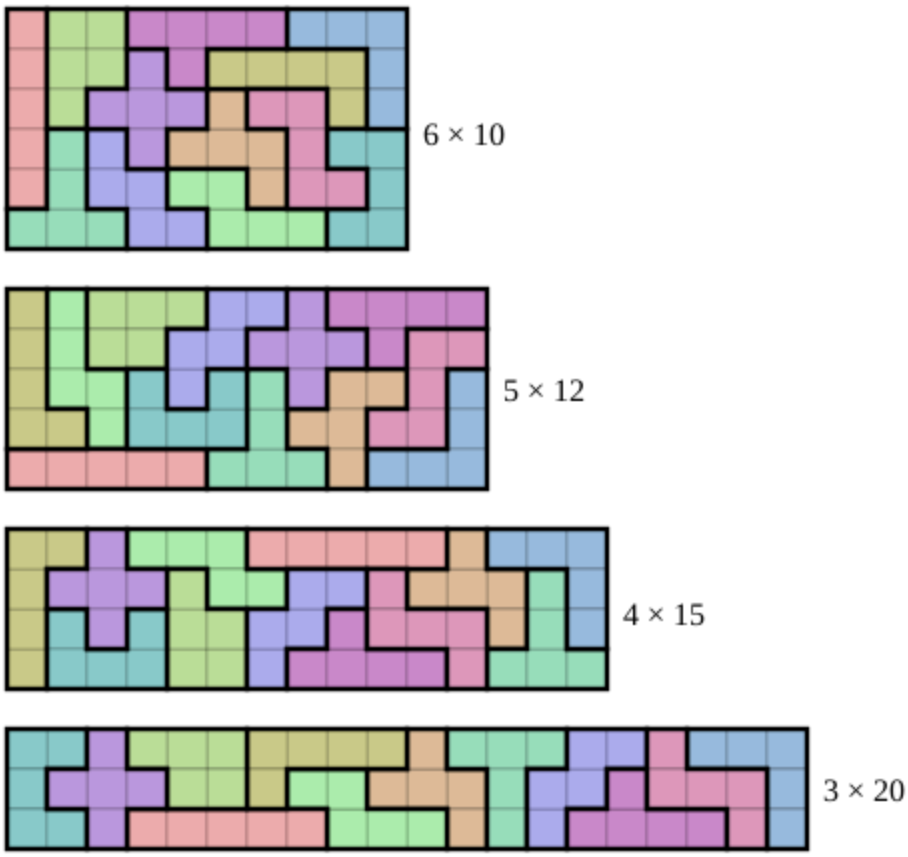
```
Z-pentomino
```

For 2D figures in general there are two more categories:

- Being orientable in 2 ways by a rotation of 90°, with two axes of reflection symmetry, both aligned with the diagonals. This type of symmetry requires at least a heptomino.
- Being orientable in 2 ways, which are each other's mirror images, for example a swastika. This type of symmetry requires at least an octomino.

Games

Tiling puzzle (2D)



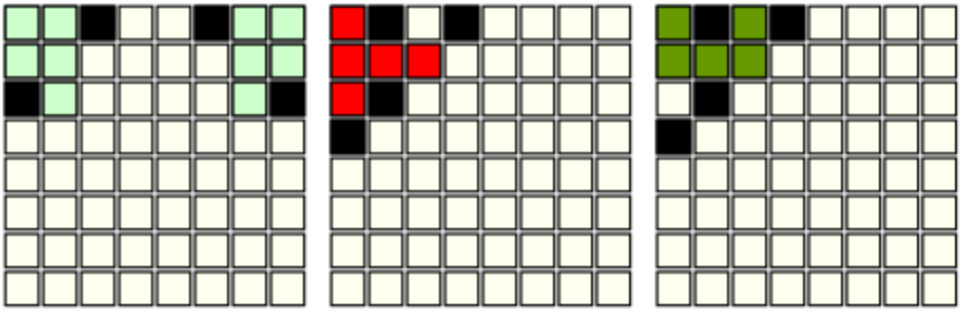
Example tilings

A standard **pentomino puzzle** is to tile a rectangular box with the pentominoes, i.e. cover it without overlap and without gaps. Each of the 12 pentominoes has an area of 5 unit squares, so the box must have an area of 60 units. Possible sizes are 6×10, 5×12, 4×15 and 3×20.

The 6×10 case was first solved in 1960 by married couple Colin Brian Haselgrove and Jenifer Haselgrove.9(https://en.wikipedia.org/wiki/Pentomino#cite_note-9) There are exactly 2,339 solutions, excluding trivial variations obtained by rotation and reflection of the whole rectangle but including rotation and reflection of a subset of pentominoes (which sometimes provides an additional solution in a simple way). The 5×12 box has 1010 solutions, the 4×15 box has 368 solutions, and the 3×20 box has just 2 solutions (one is shown in the figure, and the other one can be obtained from the solution shown by rotating, as a whole, the block consisting of the L, N, F, T, W, Y, and Z pentominoes).

A somewhat easier (more symmetrical) puzzle, the 8×8 rectangle with a 2×2 hole in the center, was solved by Dana Scott as far back as 1958.10(https://en.wikipedia.org/wiki/Pentomino#cite_note-10) There are 65 solutions. Scott's algorithm was one of the first applications of a backtracking computer program. Variations of this puzzle allow the four holes to be placed in any position. One of the external links uses this rule.

Efficient algorithms have been described to solve such problems, for instance by Donald Knuth.11(https://en.wikipedia.org/wiki/Pentomino#cite_note-11) Running on modern hardware, these pentomino puzzles can now be solved in mere seconds.

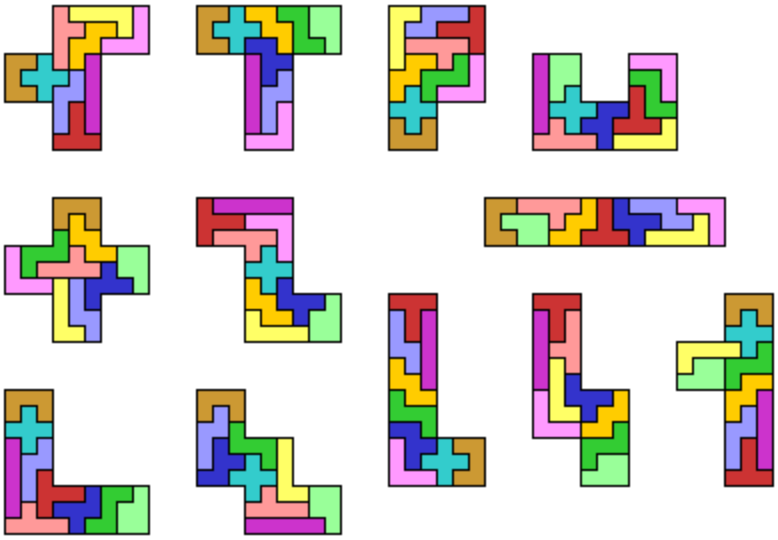


Unsolvable patterns

Most such patterns are solvable, with the exceptions of placing each pair of holes near two corners of the board in such a way that both corners could only be fitted by a P-pentomino, or forcing a T-pentomino or U-pentomino in a corner such that another hole is created.

The pentomino set is the only free polyomino set that can be packed into a rectangle, with the exception of the trivial monomino and domino sets, each of which consists only of a single rectangle.

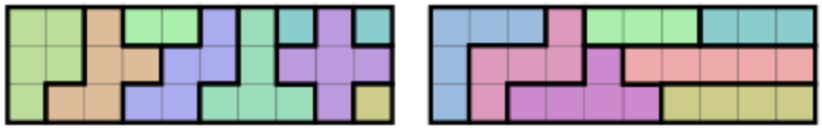
Each pentomino can be divided into nine pentominoes, as shown.



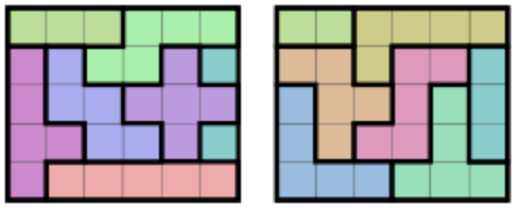
[*citation needed*]

Box filling puzzle (3D)

2 × 3 × 10 box



2 × 5 × 6 box



3 × 4 × 5 box

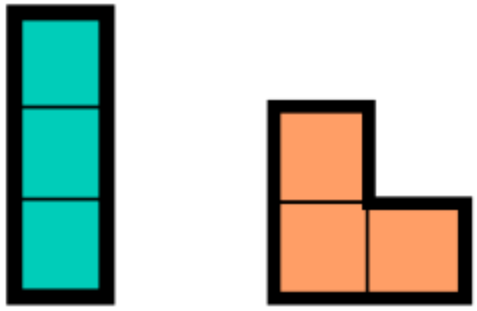
Sample solutions to pentacube puzzles of the stated dimensions, drawn one layer at a time.

A **pentacube** is a polycube of five cubes. Of the 29 one-sided pentacubes, exactly twelve pentacubes are flat (1-layer) and correspond to the twelve pentominoes extruded to a depth of one square.

A **pentacube puzzle** or 3D **pentomino puzzle**, amounts to filling a 3-dimensional box with the 12 flat pentacubes, i.e. cover it without overlap and without gaps. Since each pentacube has a volume of 5 unit cubes, the box must have a volume of 60 units. Possible sizes are 2×3×10 (12 solutions), 2×5×6 (264 solutions) and 3×4×5 (3940 solutions).12(https://en.wikipedia.org/wiki/Pentomino#cite_note-12)

Alternatively one could also consider combinations of five cubes that are themselves 3D, i.e., those which include more than just the 12 "flat" single-layer thick combinations of cubes. However, in addition to the 12 "flat" pentacubes formed by extruding the pentominoes, there are 6 sets of chiral pairs and 5 additional pieces, forming a total of 29 potential pentacube pieces, which gives 145 cubes in total (=29×5); as 145 can only be packed into a box measuring 29×5×1, it cannot be formed by including the non-flat pentominoes.

This article is about the geometric shape. For the game similar to dominoes, see Triominoes.



All possible free trominos

A **tromino** or **triomino** is a polyomino of size 3, that is, a polygon in the plane made of three equal-sized squares connected edge-to-edge.1(https://en.wikipedia.org/wiki/Tromino#cite_note-1)

Symmetry and enumeration

When rotations and reflections are not considered to be distinct shapes, there are only two different *free* trominoes: "I" and "L" (the "L" shape is also called "V").

Since both free trominoes have reflection symmetry, they are also the only two *one-sided* trominoes (trominoes with reflections considered distinct). When rotations are also considered distinct, there are six *fixed* trominoes: two I and four L shapes. They can be obtained by rotating the above forms by 90°, 180° and 270°.2(https://en.wikipedia.org/wiki/Tromino#cite_note-2)[[3]](https://en.wikipedia.org/wiki/Tromino#cite_note-3)

This article is about the mathematical polygon. For the game, see dominoes.

The single free domino

In mathematics, a **domino** is a polyomino of order 2, that is, a polygon in the plane made of two equal-sized squares connected edge-to-edge.1(https://en.wikipedia.org/wiki/Domino_\

(mathematics\)#cite_note-1) When rotations and reflections are not considered to be distinct shapes, there is only one *free* domino.

Since it has reflection symmetry, it is also the only *one-sided* domino (with reflections considered distinct). When rotations are also considered distinct, there are two *fixed* dominoes: The second one can be created by rotating the one above by 90°.2(https://en.wikipedia.org/wiki/Domino_\(mathematics\)#cite_note-2)[[3]](https://en.wikipedia.org/wiki/Domino_\(mathematics\)#cite_note-3)

In a wider sense, the term *domino* is sometimes understood to mean a tile of any shape.4(https://en.wikipedia.org/wiki/Domino_\(mathematics\)#cite_note-4)

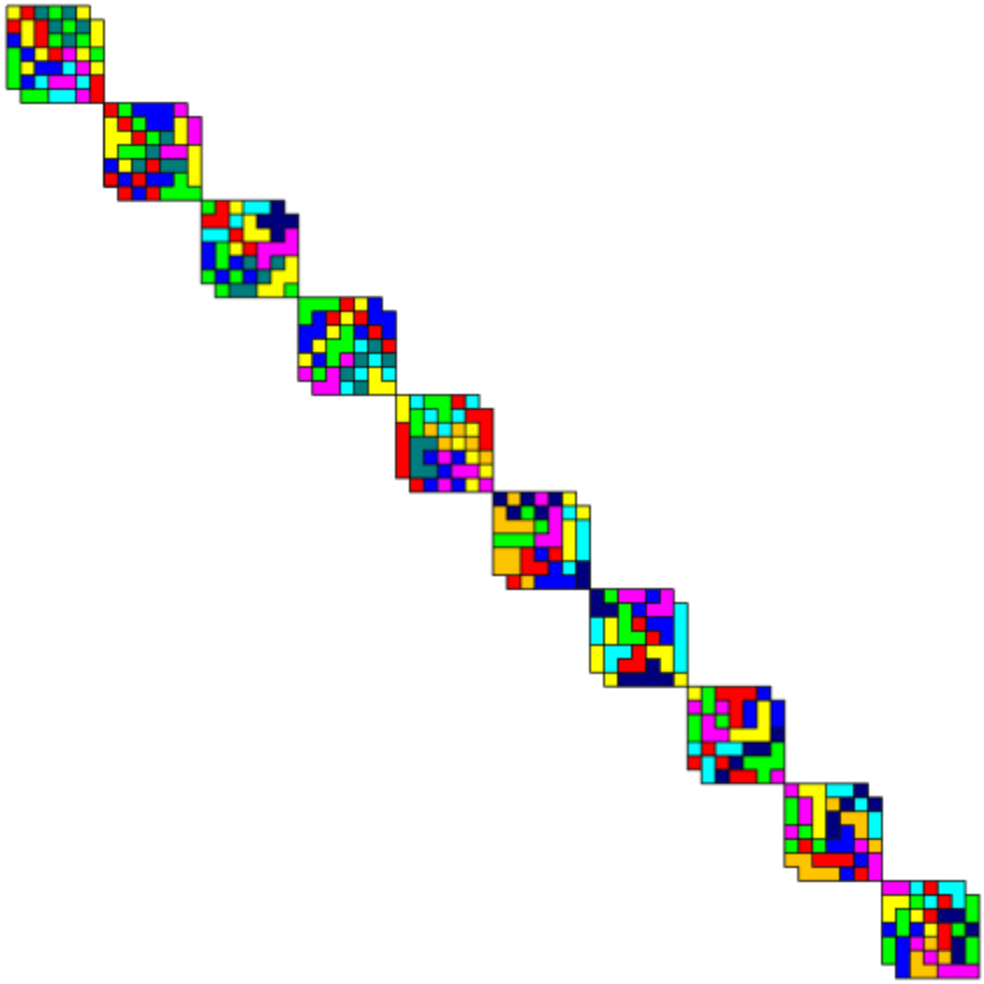
The 22 free tetrakings

A **pseudo-polyomino**, also called a **polyking**, **polyplet** or **hinged polyomino**, is a plane geometric figure formed by joining one or more equal squares edge-to-edge or corner-to-corner at 90°. It is a polyform with square cells. The polyominoes are a subset of the polykings.

The name "polyking" refers to the king in chess. The *n*-kings are the *n*-square shapes which could be occupied by a king on an infinite chessboard in the course of legal moves.

Golomb uses the term *pseudo-polyomino* referring to kingwise-connected sets of squares.1(https://en.wikipedia.org/wiki/Pseudo-polyomino#cite_note-golomb1994-1)

Enumeration of polykings



10 congruent mutilated chessboards 7x7 constructed with the 94 pseudo-pentominoes, or pentaplets

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> eBPF evolved from the classic Berkeley Packet Filter (cBPF, a retroactively-applied name). At the most basic level, it introduced the use of ten 64-bit registers (instead of two 32-bit long registers for cBPF), different jump semantics, a call instruction and corresponding register passing convention, new instructions, and a different encoding for these instructions.

Free, one-sided, and fixed polykings

There are three common ways of distinguishing polyominoes and polykings for enumeration:1(https://en.wikipedia.org/wiki/Pseudo-polyomino#cite_note-golomb1994-1)

- *free* polykings are distinct when none is a rigid transformation (translation, rotation, reflection or glide reflection) of another (pieces that can be picked up

and flipped over).

- *one-sided* polykings are distinct when none is a translation or rotation of another (pieces that cannot be flipped over).
- *fixed* polykings are distinct when none is a translation of another (pieces that can be neither flipped nor rotated).

A **tetromino** is a geometric shape composed of four squares, connected orthogonally (i.e. at the edges and not the corners).1(https://en.wikipedia.org/wiki/Tetromino#cite_note-GolombPolyominoes-1)[[2]](https://en.wikipedia.org/wiki/Tetromino#cite_note-2) Tetrominoes, like dominoes and pentominoes, are a particular type of polyomino. The corresponding polycube, called a **tetracube**, is a geometric shape composed of four cubes connected orthogonally.

A popular use of tetrominoes is in the video game *Tetris* created by the Soviet game designer Alexey Pajitnov, which refers to them as **tetriminos**.3(https://en.wikipedia.org/wiki/Tetromino#cite_note-3) The tetrominoes used in the game are specifically the one-sided tetrominoes.

Types of tetrominoes

Free tetrominoes

Polyominos are formed by joining unit squares along their edges. A free polyomino is a polyomino considered up to congruence. That is, two free polyominos are the same if there is a combination of translations, rotations, and reflections that turns one into the other. A free tetromino is a free polyomino made from four squares. There are five free tetrominoes.

The free tetrominoes have the following symmetry:

- Straight: vertical and horizontal reflection symmetry, and two-fold rotational symmetry
- Square: vertical, horizontal, and diagonal reflection symmetry, and four-fold rotational symmetry
- T: vertical reflection symmetry only
- L: no symmetry
- S: two-fold rotational symmetry only

"straight tetromino"

"square tetromino"

"T-tetromino"

"L-tetromino"

"skew tetromino"

One-sided tetrominoes

One-sided tetrominoes are tetrominoes that may be translated and rotated but not reflected. They are used by, and are overwhelmingly associated with, *Tetris*. There are seven distinct one-sided tetrominoes. These tetrominoes are named by the letter of the alphabet they most closely resemble. The "I", "O", and "T" tetrominoes have reflectional symmetry, so it does not matter whether they are considered as free tetrominoes or one-sided tetrominoes. The remaining four tetrominoes, "J", "L", "S", and "Z", exhibit a phenomenon called chirality. J and L are reflections of each other, and S and Z are reflections of each other.

As free tetrominoes, J is equivalent to L, and S is equivalent to Z, but in two dimensions and without reflections, it is not possible to transform J into L or S into Z.

I

O

T

J

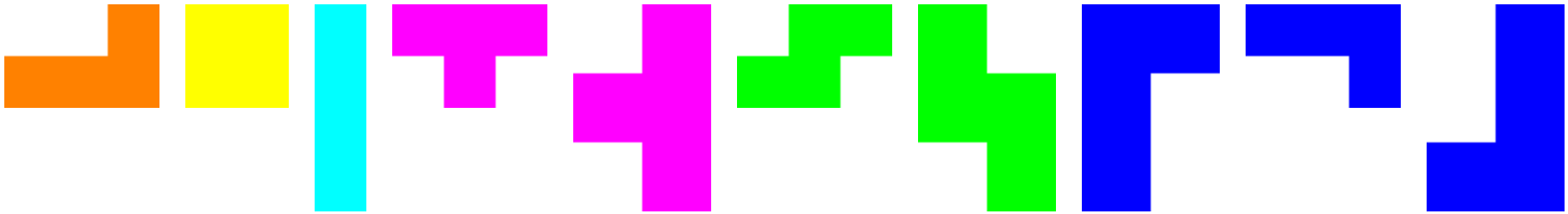
L

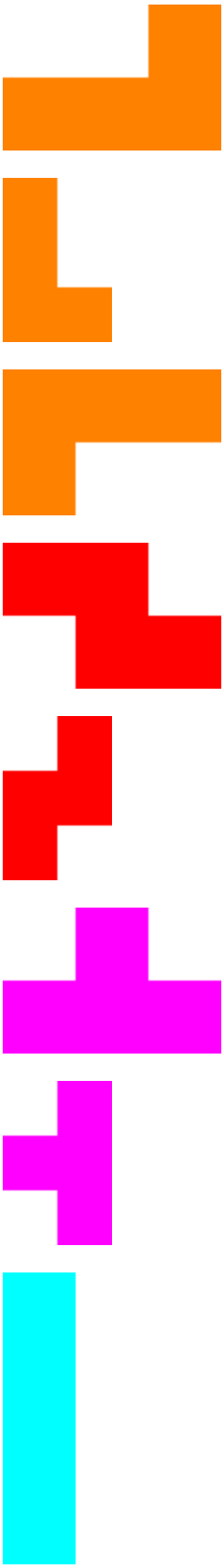
S

Z

Fixed tetrominoes

The fixed tetrominoes allow only translation, not rotation nor reflection. There are two distinct fixed I-tetrominoes, four J, four L, one O, two S, four T, and two Z, for a total of 19 fixed tetrominoes.

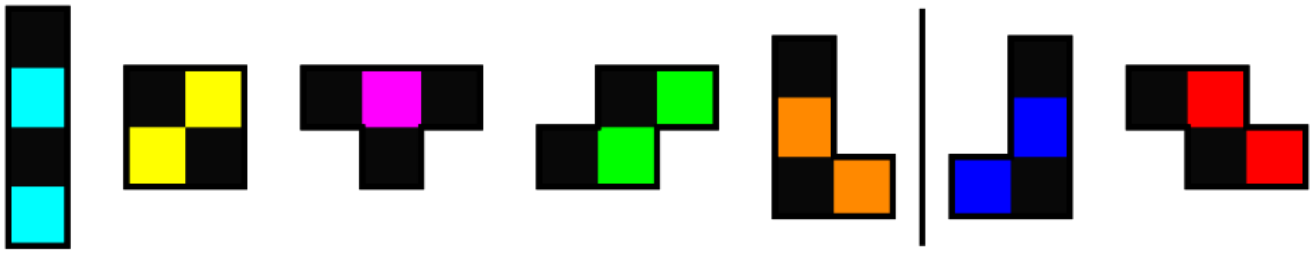




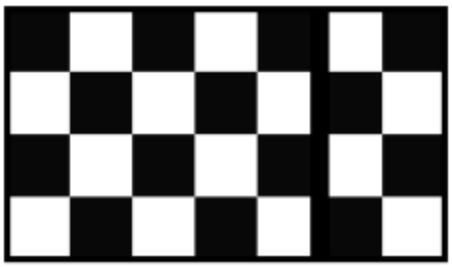
Tiling a rectangle

Filling a rectangle with one set of tetrominoes

A single set of free tetrominoes or one-sided tetrominoes cannot fit in a rectangle. This can be shown with a proof similar to the mutilated chessboard argument. A 5×4 rectangle with a checkerboard pattern has 20 squares, containing 10 light squares and 10 dark squares, but a complete set of free tetrominoes has either 11 dark squares and 9 light squares, or 11 light squares and 9 dark squares. This is due to the T tetromino having either 3 dark squares and one light square, or 3 light squares and one dark square, while all other tetrominoes each have 2 dark squares and 2 light squares. Similarly, a 7×4 rectangle has 28 squares, containing 14 squares of each shade, but the set of one-sided tetrominoes has either 15 dark squares and 13 light squares, or 15 light squares and 13 dark squares. By extension, any odd number of sets for either type cannot fit in a rectangle. Additionally, the 19 fixed tetrominoes cannot fit in a 4×19 rectangle. This was discovered by exhausting all possibilities in a computer search.



The free tetrominoes (left side of line) have 11 dark squares and 9 light squares.
The one-sided tetrominoes (all 7 shown above) have 15 dark squares and 13 light squares.



A 5×4 board has 10 squares each color.
A 7×4 board has 14 squares each color.

Parity

A further consequence of the T tetromino having 3 squares of one colour and 1 square of the other is that any rectangle containing an even number of squares must contain an even number of T tetrominoes. Conversely, any rectangles containing an odd number of squares must contain an odd number of T tetrominoes.

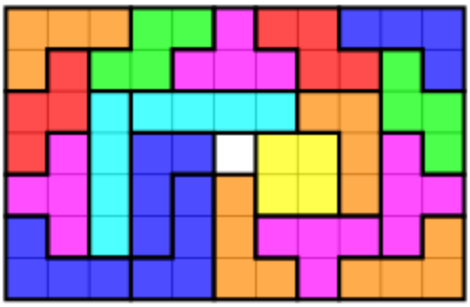
Filling a modified rectangle with one set of tetrominoes

All three sets of tetrominoes can fit rectangles with holes:

- All 5 free tetrominoes fit a 7×3 rectangle with a hole.
- All 7 one-sided tetrominoes fit a 6×5 rectangle with two holes of the same "checkerboard color".
- All 19 fixed tetrominoes fit a 11×7 rectangle with a hole.

Free tetrominoes in a rectangle with one hole

One-sided tetrominoes in a rectangle with two holes



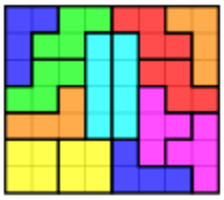
Fixed tetrominoes in rectangle with one hole

Filling a rectangle with two sets of tetrominoes

Two sets of free or one-sided tetrominoes can fit into a rectangle in different ways, as shown below:

Two sets of free tetrominoes in a 5×8 rectangle

Two sets of free tetrominoes in a 4×10 rectangle



Two sets of one-sided tetrominoes in a 8×7 rectangle

Two sets of one-sided tetrominoes in a 14×4 rectangle

Etymology

The name "tetromino" is a combination of the prefix *tetra-* 'four' (from Ancient Greek τετρα-), and "domino". The name was introduced by Solomon W. Golomb in 1953 along with other nomenclature related to polyominos.4(https://en.wikipedia.org/wiki/Tetromino#cite_note-4)[[1]](https://en.wikipedia.org/wiki/Tetromino#cite_note-GolombPolyominoes-1)

Filling a box with tetracubes

Each of the five free tetrominoes has a corresponding tetracube, which is the tetromino extruded by one unit. J and L are the same tetracube, as are S and Z, because one may be rotated around an axis parallel to the tetromino's plane to form the other. Three more tetracubes are possible, all created by placing a unit cube on the bent tricube:

I
"straight tetracube"

O
"square tetracube"

T
"T-tetracube"

L
"L-tetracube"

J is the same as L in 3D

S
"skew tetracube"

Z is the same as S in 3D

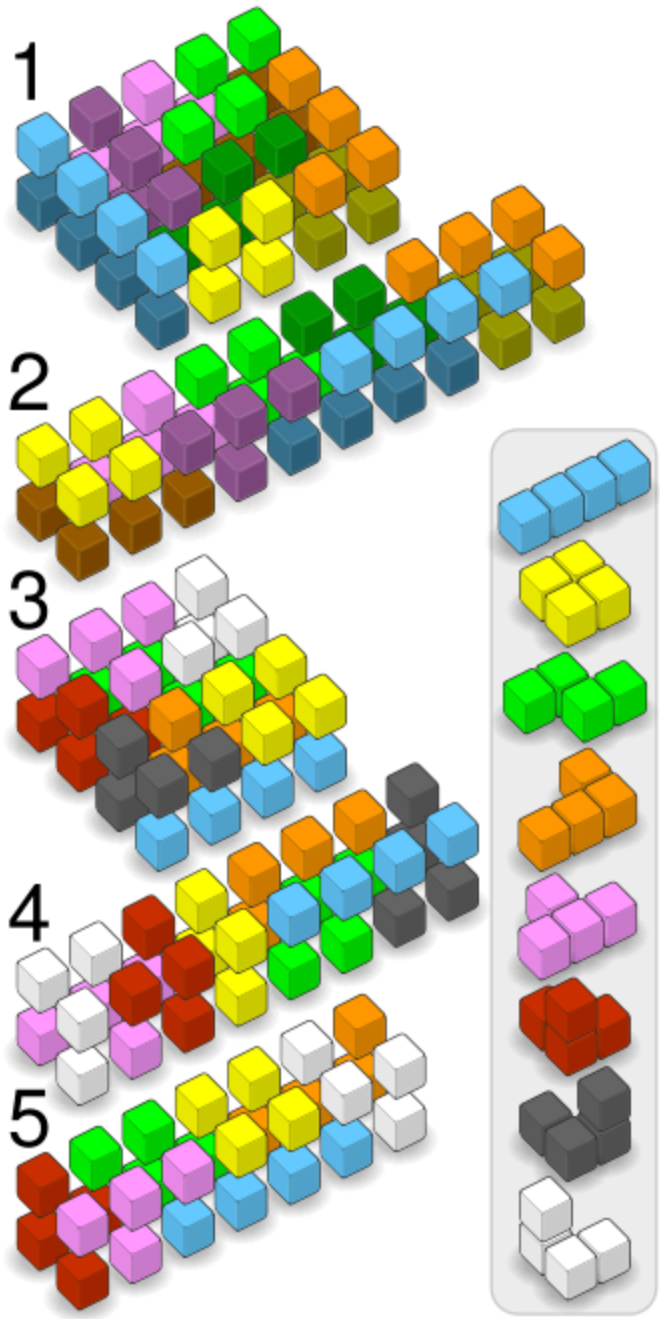
B
"Branch"

D
"Right Screw"

F
"Left Screw"

The tetracubes can be packed into two-layer 3D boxes in several different ways, based on the dimensions of the box and criteria for inclusion. They are shown in both a pictorial diagram and a text diagram. For boxes using two sets of the same pieces, the pictorial diagram depicts each

set as a lighter or darker shade of the same color. The text diagram depicts each set as having a capital or lower-case letter. In the text diagram, the top layer is on the left, and the bottom layer is on the right.



1.) 2×4×5 box filled with two sets of free tetrominoes:

Z Z T t I I T T T i
L Z Z t I I I I t i
L z z t I o o z z i
L L O O I o o O O i

2.) 2×2×10 box filled with two sets of free tetrominoes:

L L L z z Z Z T O O o o z z Z Z T T T I
L I I I I t t t O O o o i i i i t I I I

3.) 2×4×4 box filled with one set of all tetrominoes:

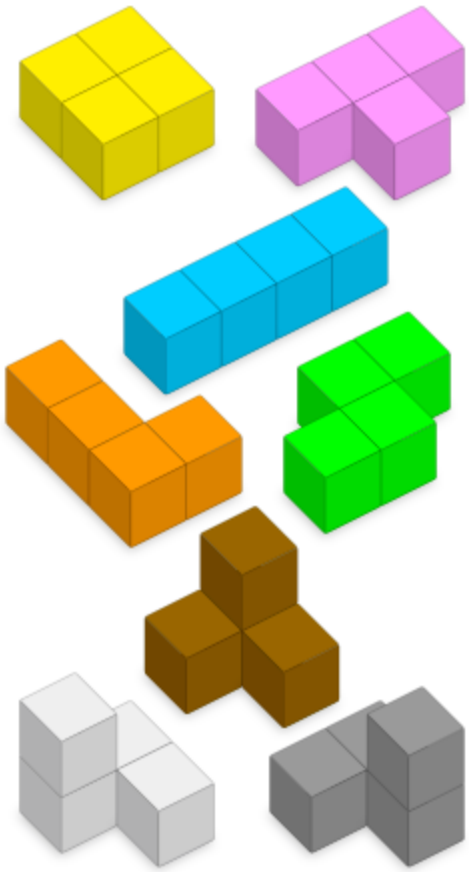
F T T T F Z Z B
F F T B Z Z B B
O O L D L L L D
O O D D I I I I

4.) 2×2×8 box filled with one set of all tetrominoes:

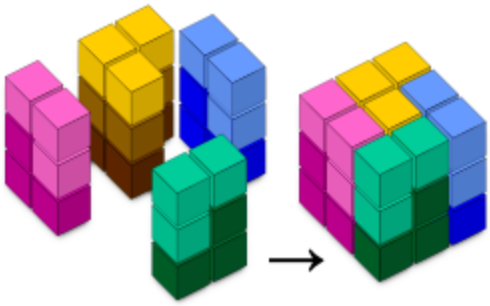
D Z Z L O T T T D L L L O B F F
D D Z Z O B T F I I I I O B B F

5.) 2×2×7 box filled with tetrominoes, with mirror-image pieces removed:

L L L Z Z B B L C O O Z Z B
C I I I I T B C C O O T T T



All 8 one-sided tetracubes – if chirality is ignored, the bottom 2 in grey are considered the same, giving 7 free tetracubes in total



A puzzle involving arranging nine L tricubes into a 3×3×3 cube

A **polycube** is an orthogonal polyhedron formed by joining one or more equal cubes face to face. Polycubes are the three-dimensional analogues of the planar polyominoes. The Soma cube, the Bedlam cube, the Diabolical cube, the Slothouber–Graatsma puzzle, and the Conway puzzle are examples of packing problems based on polycubes.1(https://en.wikipedia.org/wiki/Polycube#cite_note-1)

Enumerating polycubes

A chiral pentacube

Like polyominoes, polycubes can be enumerated in two ways, depending on whether chiral pairs of polycubes (those equivalent by mirror reflection, but not by using only translations and rotations) are counted as one polycube or two. For example, 6 tetracubes are achiral and one is chiral, giving a count of 7 or 8 tetracubes respectively.2(https://en.wikipedia.org/wiki/Polycube#cite_note-symmetry-2) Unlike polyominoes, polycubes are usually counted with mirror pairs distinguished, because one cannot turn a polycube over to reflect it as one can a polyomino given three dimensions. In particular, the Soma cube uses both forms of the chiral tetracube.

Polycubes are classified according to how many cubical cells they have:3(https://en.wikipedia.org/wiki/Polycube#cite_note-3)

n	Name of *n*-polycube	Number of one-sided *n*-polycubes (reflections counted as distinct) (sequence A000162 in the OEIS)	Number of free *n*-polycubes (reflections counted together) (sequence A038119 in the OEIS)
1	monocube	1	1
2	dicube	1	1
3	tricube	2	2
4	tetracube	8	7
5	pentacube	29	23
6	hexacube	166	112
7	heptacube	1023	607
8	octacube	6922	3811

Fixed polycubes (both reflections and rotations counted as distinct (sequence A001931 in the OEIS)), one-sided polycubes, and free polycubes have been enumerated up to *n*=22. Specific families of polycubes have also been investigated.4(https://en.wikipedia.org/wiki/Polycube#cite_note-4)[[5]](https://en.wikipedia.org/wiki/Polycube#cite_note-5)

Symmetries of polycubes

As with polyominoes, polycubes may be classified according to how many symmetries they have. Polycube symmetries (conjugacy classes of subgroups of the achiral octahedral group) were first enumerated by W. F. Lunnon in 1972. Most polycubes are asymmetric, but many have more complex symmetry groups, all the way up to the full symmetry group of the cube with 48 elements. There are 33 different symmetry types that a polycube can have (including asymmetry).2(https://en.wikipedia.org/wiki/Polycube#cite_note-symmetry-2)

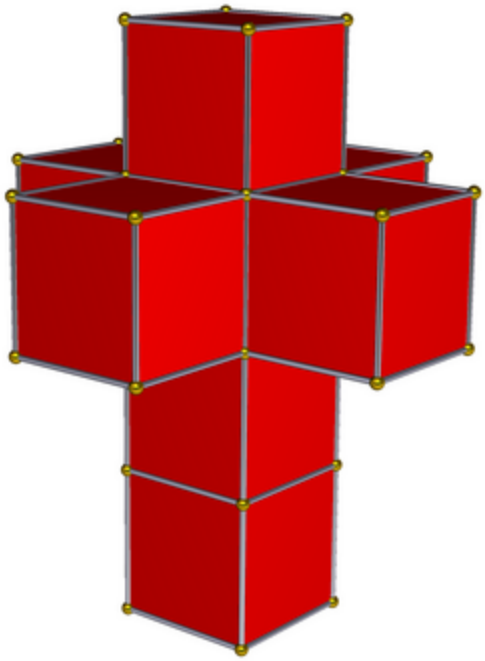
Properties of pentacubes

12 pentacubes are flat and correspond to the pentominoes. 5 of the remaining 17 have mirror symmetry, and the other 12 form 6 chiral pairs.

The bounding boxes of the pentacubes have sizes 5×1×1, 4×2×1, 3×3×1, 3×2×1, 3×2×2, and 2×2×2.6(https://en.wikipedia.org/wiki/Polycube#cite_note-6)

A polycube may have up to 24 orientations in the cubic lattice, or 48, if reflection is allowed. Of the pentacubes, 2 flats (5-1-1 and the cross) have mirror symmetry in all three axes; these have only three orientations. 10 have one mirror symmetry; these have 12 orientations. Each of the remaining 17 pentacubes has 24 orientations.

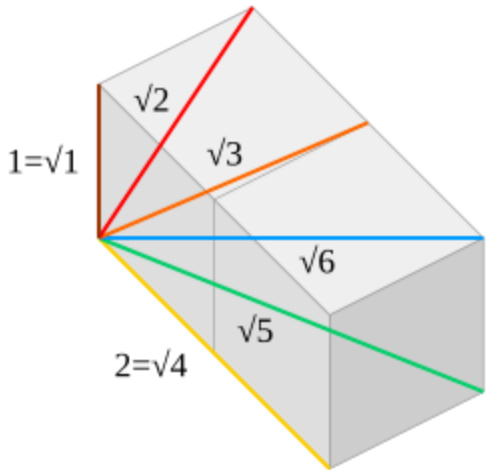
Octacube and hypercube unfoldings



The Dalí cross

The tesseract (four-dimensional hypercube) has eight cubes as its facets, and just as the cube can be unfolded into a hexomino, the tesseract can be unfolded into an octacube. One unfolding, in particular, mimics the well-known unfolding of a cube into a Latin cross: it consists of four cubes stacked one on top of each other, with another four cubes attached to the exposed square faces of the second-from-top cube of the stack, to form a three-dimensional double cross shape. Salvador Dalí used this shape in his 1954 painting *[Crucifixion (Corpus Hypercubus)](https://en.wikipedia.org/wiki/Crucifixion*(Corpus*Hypercubus) "Crucifixion (Corpus Hypercubus")*7(https://en.wikipedia.org/wiki/Polycube#cite_note-7) and it is described in Robert A. Heinlein's 1940 short story "And He Built a Crooked House".8(https://en.wikipedia.org/wiki/Polycube#cite_note-8) In honor of Dalí, this octacube has been called the *Dalí cross*.9(https://en.wikipedia.org/wiki/Polycube#cite_note-hut-9)[[10]] (https://en.wikipedia.org/wiki/Polycube#cite_note-pucc-10) It can tile space.9(https://en.wikipedia.org/wiki/Polycube#cite_note-hut-9)

More generally (answering a question posed by Martin Gardner in 1966), out of all 3811 different free octacubes, 261 are unfoldings of the tesseract.9(https://en.wikipedia.org/wiki/Polycube#cite_note-hut-9)[[11]] (https://en.wikipedia.org/wiki/Polycube#cite_note-11)



Unlike in three dimensions in which distances between vertices of a polycube with unit edges excludes √7 due to Legendre's three-square theorem, Lagrange's four-square theorem states that the analogue in four dimensions yields square roots of every natural number

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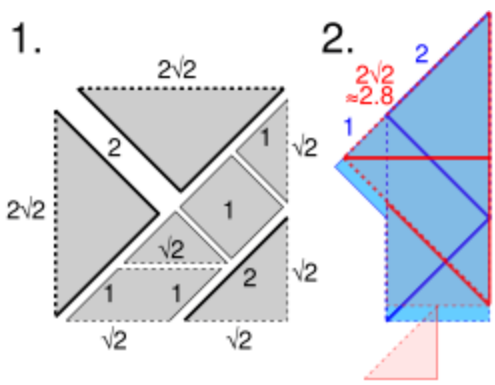
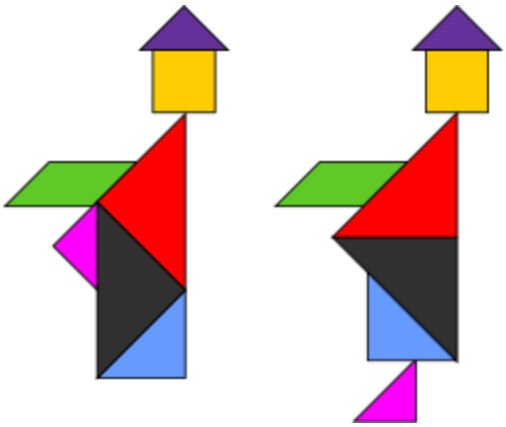
Paradoxes



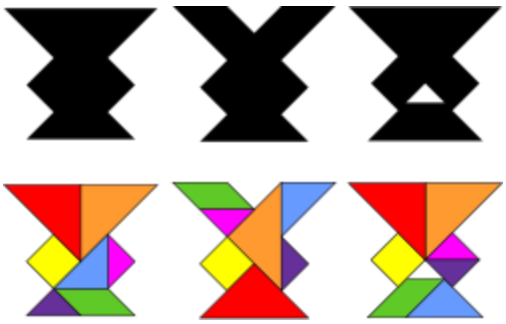
Explanation of the two-monks paradox:
In figure 1, side lengths are labelled assuming the square has unit sides.
In figure 2, overlaying the bodies shows that footless body is larger by the foot's area. The change in area is often unnoticed as √2 is close to 1.5.

A tangram paradox is a dissection fallacy: Two figures composed with the same set of pieces, one of which seems to be a proper subset of the other.21(https://en.wikipedia.org/wiki/Tangram#cite_note-mathematica-21) One famous paradox is that of the two monks, attributed to Henry Dudeney, which consists of two similar shapes, one with and the other missing a foot.22(https://en.wikipedia.org/wiki/Tangram#cite_note-dudeney-22) In reality, the area of the foot is compensated for in the second figure by a subtly larger body.

The two-monks paradox – two similar shapes but one missing a foot:



The Magic Dice Cup tangram paradox – from Sam Loyd's book *The 8th Book of Tan* (1903).23(https://en.wikipedia.org/wiki/Tangram#cite_note-eighth_book_1-23) Each of these cups was composed using the same seven geometric shapes. But the first cup is whole, and the others contain vacancies of different sizes. (Notice that the one on the left is slightly shorter than the other two. The one in the middle is ever-so-slightly wider than the one on the right, and the one on the left is narrower still.)24(https://en.wikipedia.org/wiki/Tangram#cite_note-24)



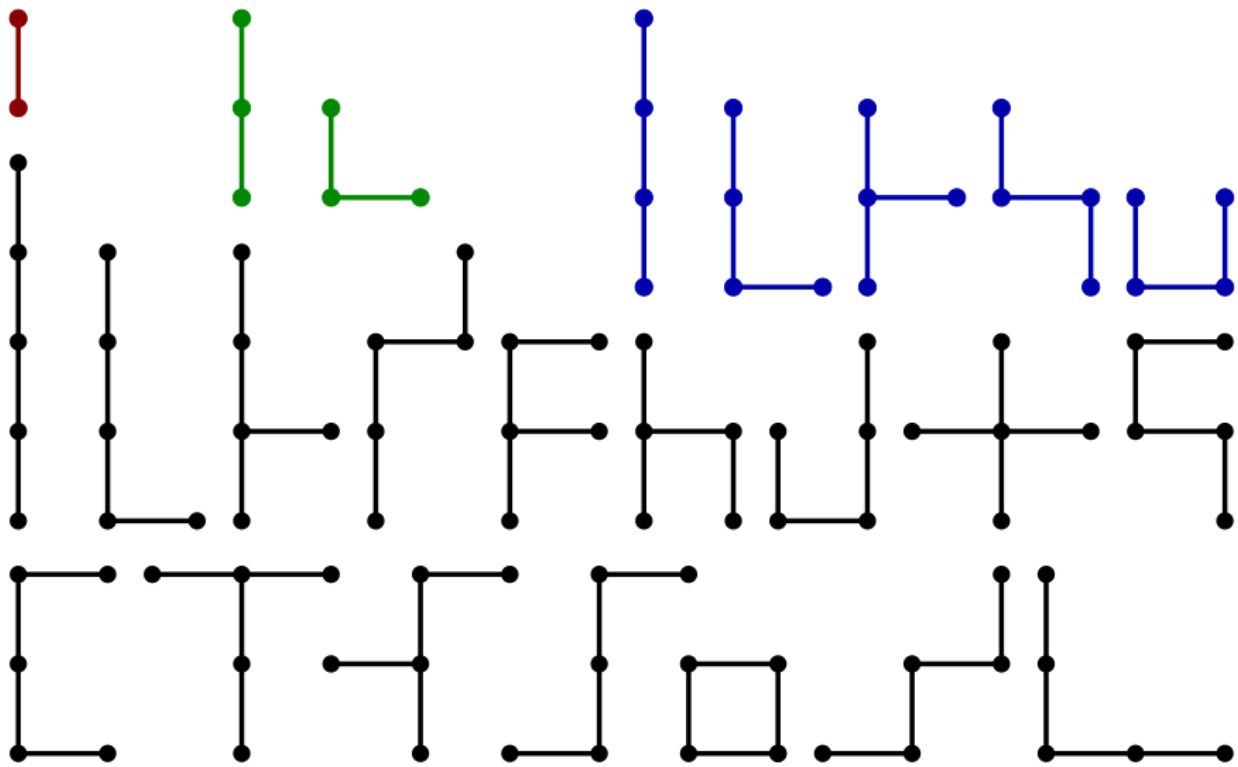
Clipped square tangram paradox – from Loyd's book *The Eighth Book of Tan* (1903):23(https://en.wikipedia.org/wiki/Tangram#cite_note-eighth_book_1-23)

> The seventh and eighth figures represent the mysterious square, built with seven pieces: then with a corner clipped off, and still the same seven pieces employed.25(https://en.wikipedia.org/wiki/Tangram#cite_note-loyd-25)

The set of *n*-sticks that contain no closed loops is equivalent, with some duplications, to the set of (*n*+1)-ominos, as each vertex at the end of every line segment can be replaced with a single

square of a polyomino. For example, the set of tristicks is equivalent to the set of Tetrominos. In general, an n-stick with m loops is equivalent to a $(n-m+1)$-omino (as each loop means that one line segment does not add a vertex to the figure).

Diagram



The free square polysticks of sizes 1 through 4, including 1 monostick (red), 2 disticks (green), 5 tristicks (blue), and 16 tetrasticks (black).

From Wikipedia, the free encyclopedia

A set of 60 attribute blocks

Attribute blocks, also called **logic blocks**, are mathematical manipulatives used to teach logic. Each block in a set has a unique combination of four attributes, namely size, color, shape, and thickness.1(https://en.wikipedia.org/wiki/Attribute_blocks#cite_note-Dienes_1976-1)